Technical Report: Multi-Label Land-Cover Recognition Using Sentinel-2 Multispectral Classification on the BigEarthNet Dataset

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Abstract

Land-cover mapping is a foundational pillar of earth observation, environmental monitoring, agricultural assessment, and spatial planning. In heterogeneous natural and anthropogenic terrains, satellite image footprints frequently encompass multiple surface categories simultaneously (e.g., agricultural fields bordered by woodlots, or rivers traversing rural settlements). Conventional single-label classification assigns an exclusive category to each scene, fundamentally misrepresenting mixed landscapes. In this project, we implement and evaluate a multi-label remote sensing classification framework using Sentinel-2 multispectral imagery from the benchmark BigEarthNet v2.0 (*reBEN*) archive.

We curated and verified a 500-patch genuine dataset originating from Sentinel-2 tile `T33UUP` in the Austrian Central Alps, strictly partitioned into official, disjoint splits: **300 Training**, **100 Validation**, and **100 Test** patches. We retained six strategically chosen diagnostic spectral bands: Blue ($B02$), Green ($B03$), Red ($B04$), Near-Infrared ($B08$), and Shortwave-Infrared ($B11, B12$). The 20 m SWIR bands were harmonized to 10 m (120×120 pixels) using bilinear interpolation, and all bands were radiometrically converted to physical Bottom-Of-Atmosphere (BOA) surface reflectance in $[0.0, 1.0]$.

We trained a regularized 4-block Convolutional Neural Network (`MultispectralCNN`, ~617k parameters) under `BCEWithLogitsLoss` and conducted a controlled **Band Ablation Experiment** comparing:

- **M_1 (Visible RGB):** Bands $B02, B03, B04$ (3 input channels)
- **M_2 (Visible + NIR):** Bands $B02, B03, B04, B08$ (4 input channels)
- **M_3 (Full Multispectral):** Bands $B02, B03, B04, B08, B11, B12$ (6 input channels)

At the standard decision threshold ($T=0.5$), ***$M2$ achieved the highest overall performance***, *recording a test* ***Macro-F1 of 0.7110***, ***Micro-F1 of 0.8205***, *and* ***Exact Match subset accuracy of 38.0%***, *significantly outperforming baseline $M1$* (**Macro-F1 of 0.5978**, **Micro-F1 of 0.7403**,

Exact Match 12.0%). Adding NIR ($B08$) provided an immediate **+11.32 percentage point gain** in Macro-F1, critically resolving canopy-density confusion (*Shrub/Transitional Vegetation* F1 surged from 0.3529 to 0.9114) and sharply elevating *Water* F1 from 0.7429 to 0.8947 with 100% precision. Conversely, **$M*3$ underperformed $M*2$**, recording a test **Macro-F1 of 0.5673**, **Micro-F1 of 0.7081**, and **Exact Match ratio of 5.0%** ($T=0.5$). When decision thresholds were calibrated on the validation set, test Macro-F1 values were **0.7076 ($M*1$)**, **0.7020 ($M*2$)**, and **0.6559 (M_3)**. On our 300-patch training split, adding two 20 m resampled SWIR channels expanded input dimensionality without sufficient sample support, leading to optimization instability and severe falsepositive over-prediction for *Grassland/Pasture* (57 false positives; precision 40.0%). Furthermore, all three models recorded **0.0000 F² on Wetland** at $T=0.5$ (0 true positives out of 27 test instances) because only 8 positive Wetland patches existed in the training split (2.7% prevalence). We discuss these findings in terms of remote sensing physics, small-sample dimensionality trade-offs, and multi-label threshold calibration.

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1. Introduction

1.1 The Imperative for Satellite Land-Cover Mapping

Automated, high-fidelity land-use and land-cover (LULC) identification from spaceborne platforms is essential for quantifying environmental change, assessing crop phenology, modeling hydrological basins, and monitoring deforestation. Modern multispectral instruments, notably the European Space Agency's (ESA) Sentinel-2 constellation, provide global systematic coverage with 5-day revisit frequencies, offering 10 m and 20 m spatial resolutions across visible, near-infrared (NIR), and shortwave-infrared (SWIR) wavelengths.

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[1] .2 Limitations of Single-Label Classification

In conventional remote sensing classification, each observation unit (pixel, polygon, or spatial tile) is forced into a single mutually exclusive category: $$y \in {1, 2, \dots, K}$$ However, at standard satellite patch footprints ($1.2\text{ km} \times 1.2\text{ km}$, corresponding to 120×120 pixels at 10 m resolution), genuine landscapes are intrinsically heterogeneous mosaics. A single patch frequently contains cultivated arable fields bordering woodlots, alpine pastures framing stream channels, and rural built-up settlements. Single-label classification arbitrarily suppresses subordinate classes or fails on mixed pixels.

[2] .3 Multi-Label Formulation

To model landscape complexity without artificial mutual exclusivity, the ground truth for each patch is structured as a binary multi-hot vector: $$\mathbf{y} = [y*1, y*2, \dots, y_K]^\top \in {0, 1}^K$$ where $y*k = 1$ denotes the confirmed presence of land-cover category k, and $y*k = 0$ denotes its absence. Under this formulation, multiple surface types can legitimately evaluate to 1 simultaneously. In our curated dataset, **82.0% of patches contain ≥ 2 co-occurring classes** (mean: 3.08 classes per patch).

1.4 Central Research Questions

1. *Can visible, NIR, and SWIR observations accurately identify multiple co-occurring surface types in complex landscapes?*
2. *Does adding NIR ($B08$) and SWIR ($B11, B12$) consistently improve classification accuracy over standard visible RGB, and how does spectral dimensionality interact with small training sample sizes?*

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2. Dataset and Spectral Bands

2.1 The BigEarthNet v2.0 (*reBEN*) Dataset

We utilize the open-access BigEarthNet v2.0 benchmark archive (Clasen et al., 2024; Sumbul et al., 2019, 2021; bigearth.net), released by the Remote Sensing Image Analysis (RSiM) and DIMA groups at Technische Universität Berlin. BigEarthNet v2.0 represents a major scientific upgrade over v1.0: • **Atmospheric Correction:** All Sentinel-2 tiles are processed to Level-2A Bottom-Of-Atmosphere (BOA) surface reflectance via `sen2cor v2.11` .

- **Label Integrity:** Ground truth labels were re-extracted using the CORINE Land Cover 2018 database (`CLC2018 v2020_u1`).
- **Official Splits:** Partitions are assigned across geographic tiles to prevent spatial autocorrelation leakage.

2.2 Geographic Provenance & Dataset Partitions

To enable rigorous, fully reproducible model training and Colab compatibility without downloading the prohibitive 59 GiB archive, we curated a genuine 500-patch dataset from the first tile in BigEarthNet: **Tile** `T33UUP` , situated in the **Austrian Central Alps** (UTM Zone 33N).

- **Landscape Context:** Alpine and sub-alpine topography (coniferous and mixed forests, managed pastures, agricultural valleys, settlements, mountain streams).
- **Official Split Assignments:**
  - **Training Split:** 300 patches (60.0%)
  - **Validation Split:** 100 patches (20.0%)
  - **Test Split:** 100 patches (20.0%)
- **Zero Leakage:** Strictly disjoint partitions ($\text{Train} \cap \text{Val} = 0$, $\text{Train} \cap \text{Test} = 0$, $\text{Val} \cap \text{Test} = 0$).
- **Inspection Patch Protocol:** All 5 Phase 1 inspection patches (`T33UUP_26_57` , `T33UUP_27_55` , `T33UUP_27_56` , `T33UUP_27_57` , `T33UUP_27_58`) reside strictly in the Test split.
- **Single-Tile Geographic Scope:** Because all 500 patches originate from tile `T33UUP` , this study evaluates spectral ablation and multi-label modeling under regional conditions; it does not test intertile or cross-continental transfer.

2.3 Retained Sentinel-2 Diagnostic Bands

Band	Spectral Region	Central λ	Bandwidth	Native Res.	Grid Shape (1.2 km)	Primary Spectroradiometric Role
B02	Blue	490 nm	65 nm	10 m	120×120	Atmospheric penetration; water body clarity; soil vs vegetation contrast
B03	Green	560 nm	35 nm	10 m	120×120	Chlorophyll green reflectance peak; general vegetative vigor
B04	Red	665 nm	30 nm	10 m	120×120	Chlorophyll-a absorption maximum; defines base of red-edge
B08	NearInfrared (NIR)	842 nm	115 nm	10 m	120×120	Leaf spongy mesophyll scattering (vegetation plateau); water absorption
B11	Shortwave Infrared-1	1610 nm	90 nm	20 m	$60 \times 60 \to 120 \times 120$	Foliar moisture absorption; soil moisture; built-up roof differentiation
B12	Shortwave Infrared-2	2190 nm	180 nm	20 m	$60 \times 60 \to 120 \times 120$	Mineral background; dry non-photosynthetic biomass; soil silica

Input tensors are structured as $\mathbf{X} \in \mathbb{R}^{C \times 120 \times 120}$, where $C \in \{3, 4, 6\}$.

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3. Seven-Class Land-Cover Scheme and CORINE Label Mapping

The 19 official CORINE Land Cover (CLC) classes in BigEarthNet v2.0 are mapped into **7 broad canonical classes**:

```
Col 0: Forest
  Col 1: Agriculture
  Col 2: Grassland/Pasture
  Col 3: Built-up/Artificial Surfaces
  Col 4: Water
  Col 5: Wetland
  Col 6: Shrub/Transitional Vegetation
```

Complete CORINE Land Cover Mapping Table

Original CORINE Land Cover Label	Mapped Broad Class	Status	Spectroradiometric & Ecological Rationale
Broad-leaved forest	**Forest**	Mapped	Deciduous broad-leaved tree canopy cover $> 30\%$.
Coniferous forest	**Forest**	Mapped	Evergreen needle-leaved tree canopy cover $> 30\%$.
Mixed forest	**Forest**	Mapped	Mosaic where neither broad-leaved nor needle-leaved dominates.
Arable land	**Agriculture**	Mapped	Regularly cultivated parcels for annual crop production.
Permanent crops	**Agriculture**	Mapped	Vineyards, orchards, fruit trees, and berry plantations.
Complex cultivation patterns	**Agriculture**	Mapped	Small cultivated parcels interspersed with rural dwellings.
Land principally occupied by agriculture, natural veg.	**Agriculture**	Mapped	Farmland interspersed with significant natural woodland patches.
Agro-forestry areas	**Agriculture**	Mapped	Annual crops or pasture combined with forest trees.
Pastures	**Grassland/Pasture**	Mapped	Managed perennial herbaceous cover grazed by livestock.
Natural grassland and sparsely vegetated areas	**Grassland/Pasture**	Mapped	Low-productivity alpine meadows, open steppe, rocky grasslands.
Urban fabric	**Built-up/Artificial Surfaces**	Mapped	Continuous and discontinuous residential, commercial, civic fabric.

Industrial or commercial units	**Built-up/Artificial Surfaces**	Mapped	Factories, industrial facilities, infrastructure, transport nodes.
Inland waters	**Water**	Mapped	Freshwater lakes, reservoirs, canals, and river channels.
Marine waters	**Water**	Mapped	Coastal waters, estuaries, bays, and open seas.
Inland wetlands	**Wetland**	Mapped	Freshwater marshes, peatbogs, seasonally waterlogged terrain.
Coastal wetlands	**Wetland**	Mapped	Salt marshes, salines, and tidal flats.
Transitional woodland, shrub	**Shrub/Transitional Vegetation**	Mapped	Bushy regrowth, young saplings, clearcut regeneration, maquis.
Moors, heathland and sclerophyllous vegetation	**Shrub/Transitional Vegetation**	Mapped	Low woody scrub, heathland, gorse, and dwarf alpine shrubs.
Original CORINE Land Cover Label	**Mapped Broad Class**	**Status**	**Spectroradiometric & Ecological Rationale**
Beaches, dunes, sands	**[None - Excluded]**	**Excluded**	Bare coastal sand/shingle lacking vegetation (0.27% of archive, 0 in $T33UUP$).

Audit Verification: The code does not silently reassign unmapped labels. Unmapped labels contribute zero 1s to target vectors. In tile `T33UUP`, *Beaches, dunes, sands* occurs 0 times.

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4. Reflectance Preprocessing & Harmonization

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1. **Bilinear Spatial Resampling:**

   Bands $B11$ and $B12$ have a native resolution of 20 m (60×60 pixels). We resample them to the 10 m working grid (120×120 pixels) using **bilinear interpolation** (`PIL.Image.BILINEAR`, order=1). Bilinear interpolation computes distanceweighted averages of four adjacent pixel centers, generating smooth, continuous spatial gradients without ringing artifacts (unlike bicubic) or artificial blocky steps (unlike nearest neighbor).

2. **Radiometric Scaling to Surface Reflectance:**

   In accordance with the ESA Sentinel-2 L2A product specification, surface reflectance ρ is quantified as $\text{DN} = 10,000 \times \rho$. We apply: $$\rho = \text{clip}\left(\frac{\text{DN}}{10,000.0}, 0.0, 1.0\right)$$ Division by 10,000 converts raw DN to Bottom-Of-Atmosphere surface reflectance. Clipping to $[0.0, 1.0]$ handles rare specular glint

highlights (e.g., solar reflections on glass roofs, greenhouse panels, or calm water surfaces $> 10,000\text{ DN}$, which represent $< 0.0042\%$ of all pixels).

3. **Preservation of Raw Data:**

   All raw GeoTIFF files remain completely unmodified in `data/curated_patches/` . Preprocessed tensors are saved in compressed archive `data/curated_dataset_phase2_1.npz` (76.66 MB).

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5. Exploratory Data Analysis & Class Frequencies

Multi-Label Complexity

- Minimum active labels per patch: **1**
- Maximum active labels per patch: **5**
- Mean active labels per patch: **3.08**
- **82.0% of all patches contain ≥ 2 co-occurring classes**, confirming that single-label classification is fundamentally inadequate for European landscapes.

Class Prevalence Across Official Splits ($N=500$)

Index	Target Class	Train ($N=300$)	Val ($N=100$)	Test ($N=100$)	Total ($N=500$)	Dataset Prevalence
0	**Forest**	258 (86.0%)	84 (84.0%)	95 (95.0%)	437	87.4%
1	**Agriculture**	142 (47.3%)	74 (74.0%)	52 (52.0%)	268	53.6%
2	**Grassland/Pasture**	209 (69.7%)	75 (75.0%)	39 (39.0%)	323	64.6%
3	**Built-up/Artificial**	130 (43.3%)	58 (58.0%)	31 (31.0%)	219	43.8%
4	**Water**	41 (13.7%)	28 (28.0%)	21 (21.0%)	90	18.0%
5	**Wetland**	**8 (2.7%)**	**16 (16.0%)**	**27 (27.0%)**	**51**	**10.2%**
6	**Shrub/Transitional**	**59 (19.7%)**	**13 (13.0%)**	**42 (42.0%)**	**114**	**22.8%**

Co-occurrence Highlights: The most frequent pairwise co-occurrences in tile `T33UUP` are **Forest + Grassland/Pasture** (wooded valley margins), **Forest + Agriculture**, and **Agriculture + Built-up** (rural villages).

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6. Multi-Label Deep Learning Methodology

Mathematical Formulation

Given an input multispectral cube $\mathbf{X} \in \mathbb{R}^{C \times 120 \times 120}$, the neural network computes unconstrained raw logits $\mathbf{z} = [z1, z2, \dots, z_K]^\top \in \mathbb{R}^7$. Unlike multi-class classification which employs a softmax activation ($\sum*k pk = 1$), multi-label classification treats each class as an independent Bernoulli trial. Independent class probabilities $\hat{p}_k \in [0, 1]$ are obtained via the standard logistic sigmoid function: $$\hat{p}*k = *sigma(*zk) = \frac{1}{1 + e^{-z_k}}$$

Loss Function

The model is optimized using **Binary Cross-Entropy with Logits** (`BCEWithLogitsLoss`), combining the sigmoid operation with binary cross-entropy: $$\mathcal{L}*{*text{BCE}}(*mathbf{z}, *mathbf{y}) = -*frac{1}{K} *sum{k=1}^K \left[y*k *cdot *log *sigma(*zk) + (1 - y*k) *cdot *log(*1 - *sigma(*zk)) \right]$$

Binary Decision Thresholding

Binary predictions $\hat{y}*k *in {*0, *1}$ *are formed by applying a decision threshold vector* $*mathbf{T} = [*T1, \dots, T_7]$: $$\hat{y}*k = *mathbb{I}(*hat{p}k \ge T_k)$$ We evaluate:

1. **Standard Default Threshold ($T_k = 0.50$):** Canonical baseline for unweighted cross-entropy.
2. **Validation-Calibrated Thresholds (T_k^*):** *Selected by grid search over $T *in [0.20, 0.80]$ on the validation split to maximize validation Macro-F1. *Crucially, test labels are never accessed during calibration.*

CNN Architecture (`MultispectralCNN`)

Implemented in `src/model.py` , the network extracts spatial-spectral features without excessive parameterization:

- **Block 1 (120×120 → 60×60):** $\text{Conv2d}(C \to 32, 3 \times 3) \to \text{BN} \to \text{ReLU} \to \text{Conv2d}(32 \to 32, 3 \times 3) \to \text{BN} \to \text{ReLU} \to \text{MaxPool2d}(2, 2)$
- **Block 2 (60×60 → 30×30):** $\text{Conv2d}(32 \to 64, 3 \times 3) \to \text{BN} \to \text{ReLU} \to \text{Conv2d}(64 \to 64, 3 \times 3) \to \text{BN} \to \text{ReLU} \to \text{MaxPool2d}(2, 2)$
- **Block 3 (30×30 → 15×15):** $\text{Conv2d}(64 \to 128, 3 \times 3) \to \text{BN} \to \text{ReLU} \to \text{Conv2d}(128 \to 128, 3 \times 3) \to \text{BN} \to \text{ReLU} \to \text{MaxPool2d}(2, 2)$
- **Block 4 (15×15 → 7×7):** $\text{Conv2d}(128 \to 256, 3 \times 3) \to \text{BN} \to \text{ReLU} \to \text{MaxPool2d}(2, 2)$
- **Global Pooling & Head:** $\text{AdaptiveAvgPool2d}((1, 1)) \to \text{Flatten} \to \text{Dropout} (0.4) \to \text{Linear}(256, 128) \to \text{ReLU} \to \text{Dropout}(0.3) \to \text{Linear}(128, 7)$ • **Trainable Parameters:** $M*1$: 616,679; $M*2$: 616,967; M_3: 617,543.

Training Configuration

- **Optimizer:** `AdamW` ($\text{lr} = 1.0 \times 10^{-3}$, $\text{weight_decay} = 1.0 \times 10^{-4}$)
- **LR Scheduler:** `CosineAnnealingLR` ($T_{max} = 35$, $\eta_{\min} = 1.0 \times 10^{-5}$)
- **Batch Size:** 16 (19 batches/epoch)
- **Early Stopping:** Monitored on validation loss (patience = 8 epochs)
- **Seed Policy:** `seed=42` across Python, NumPy, and PyTorch

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7. Experimental Results

The models were trained and evaluated using the verified metrics stored in `outputs/metrics/phase3_model_comparison.csv` and `outputs/metrics/phase3_per_class_metrics.csv`.

7.1 Overall Performance Summary on Test Set ($N=100$)

Evaluation Protocol	Model Identifier	Macro-F1	Micro-F1	Macro-Precision	Macro-Recall	Exact Match Ratio	Hamming Loss
Standard ($T=0.50$)	**M_1 (Visible RGB)**	0.5978	0.7403	0.7317	0.5588	12.0%	0.2014
Standard ($T=0.50$)	**M_2 (Visible + NIR)**	**0.7110**	**0.8205**	**0.7326**	**0.7036**	**38.0%**	**0.1500**
Standard ($T=0.50$)	**M_3 (Multispectral)**	0.5673	0.7081	0.6894	0.5666	5.0%	0.2486
Validation-Tuned	M_1 (Tuned Thresholds)	0.7076	0.8024	0.6648	0.7868	9.0%	0.1857
Validation-Tuned	M_2 (Tuned Thresholds)	0.7020	0.7928	0.6566	0.7886	7.0%	0.1971
Validation-Tuned	M_3 (Tuned Thresholds)	0.6559	0.7692	0.5939	0.7645	6.0%	0.2229

Validation-Tuned Threshold Vectors:

- M_1: $[\text{Fo}: 0.30, \text{Ag}: 0.20, \text{Gr}: 0.40, \text{Bu}: 0.40, \text{Wa}: 0.30, \text{We}: 0.50, \text{Sh}: 0.20]$
- M_2: $[\text{Fo}: 0.20, \text{Ag}: 0.20, \text{Gr}: 0.35, \text{Bu}: 0.30, \text{Wa}: 0.30, \text{We}: 0.50, \text{Sh}: 0.20]$
- M_3: $[\text{Fo}: 0.45, \text{Ag}: 0.30, \text{Gr}: 0.45, \text{Bu}: 0.45, \text{Wa}: 0.20, \text{We}: 0.50, \text{Sh}: 0.20]$

7.2 Complete Per-Class Metrics Table at Threshold $T=0.50$

Model	Class Name	Support	Precision	Recall	F1-Score	TP	FP	FN	TN
M_1	Forest	95	0.9588	0.9789	0.9688	93	4	2	1
M_1	Agriculture	52	0.9444	0.6538	0.7727	34	2	18	46
M_1	Grassland/Pasture	39	0.6604	0.8974	0.7609	35	18	4	43
M_1	Built-up/Artificial	31	0.6296	0.5484	0.5862	17	10	14	59
M_1	Water	21	0.9286	0.6190	0.7429	13	1	8	78
M_1	**Wetland**	**27**	**0.0000**	**0.0000**	**0.0000**	**0**	**0**	**27**	**73**
M_1	**Shrub/Transitional**	**42**	**1.0000**	**0.2143**	**0.3529**	**9**	**0**	**33**	**58**
M_2	Forest	95	0.9688	0.9789	0.9738	93	3	2	2
M_2	Agriculture	52	0.9512	0.7500	0.8387	39	2	13	46
M_2	Grassland/Pasture	39	0.6727	0.9487	0.7872	37	18	2	43
M_2	Built-up/Artificial	31	0.5625	0.5806	0.5714	18	14	13	55
M_2	Water	21	**1.0000**	**0.8095**	**0.8947**	**17**	**0**	**4**	**79**
M_2	**Wetland**	**27**	**0.0000**	**0.0000**	**0.0000**	**0**	**0**	**27**	**73**
M_2	**Shrub/Transitional**	**42**	**0.9730**	**0.8571**	**0.9114**	**36**	**1**	**6**	**57**
M_3	Forest	95	0.9592	0.9895	0.9741	94	4	1	1
M_3	Agriculture	52	0.9000	0.6923	0.7826	36	4	16	44
M_3	Grassland/Pasture	39	**0.4000**	0.9744	**0.5672**	38	**57**	1	4
M_3	Built-up/Artificial	31	0.5667	0.5484	0.5574	17	13	14	56
M_3	Water	21	1.0000	**0.2857**	**0.4444**	6	0	**15**	79
M_3	**Wetland**	**27**	**0.0000**	**0.0000**	**0.0000**	**0**	**0**	**27**	**73**
M_3	**Shrub/Transitional**	**42**	1.0000	0.4762	0.6452	20	0	22	58

8. Discussion of NIR and SWIR Results

8.1 The Physical Impact of NIR ($B08$): Spongy Mesophyll Scattering

Adding Band $B08$ (842 nm) provided an immediate **+11.32 percentage point increase** in default-threshold Macro-F1 (0.5978 to 0.7110) and more than tripled the Exact Match subset accuracy from **12.0% to 38.0%**.

1. **Shrub/Transitional Vegetation Breakthrough:**

In visible RGB ($M*1$), young saplings, clear-cut regrowth, and heath shrubs display chlorophyll absorption identical to closed mature forest, yielding an F1 of only 0.3529 (33 false negatives). In NIR, however, the multi-tiered architecture of open scrub scatters radiation differently from dense forest canopies with deep structural shadowing. Consequently, **$M2$ recovered 36 of 42 shrub instances**, elevating F1 to **0.9114 (+55.85 pp)**.

2. **Hydrological Delineation (Water):**

   Liquid water strongly absorbs near-infrared radiation. Pure water bodies exhibit near-zero reflectance at 842 nm. Adding $B08$ allowed M_2 to attain **100% precision on water** (0 false alarms) and boosted recall from 61.9% to 80.95%, increasing F1 from **0.7429 to 0.8947 (+15.18 pp)**.

8.2 The Impact of SWIR ($B11, B12$): Small-Sample Dimensionality Trade-Off

In broad continental benchmarks ($> 100,000$ patches), SWIR typically aids classification. However, on our 300-patch training split, **$M3$ underperformed $M2$** (Macro-F1 0.5673 vs 0.7110):

1. **Curse of Dimensionality on Small Samples:** Moving from 4 channels ($M2$) to 6 channels ($M3$) expanded the input space by 50% without additional training data, causing optimization instability.
2. **Spatial Resampling Uncertainty:** $B11$ and $B12$ originate at 20 m native resolution. Bilinear interpolation smooths pixel grids but cannot synthesize genuine 10 m spatial edges.
3. **Severe Grassland Over-Prediction:** M_3 generated **57 False Positives** on Grassland/Pasture (predicting pasture in 95 of 100 test patches), collapsing precision to 40.0% and F1 to 0.5672. Furthermore, water recall dropped to 28.57% (15 omissions).

Scientific Takeaway: This experiment does not establish that SWIR is universally unhelpful; rather, it demonstrates that without pretraining or large-scale regularization, increasing spectral dimensionality on small datasets can induce overfitting.

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9. Error Analysis and Wetland Classification Failure

9.1 Diagnostic Breakdown of Test Predictions

At the standard $T=0.50$ threshold:

- **Exact Match Ratio: 38.0%** in $M2$, **12.0% in $M1$**, and **5.0%** in M_3. Because exact match requires all 7 binary decisions to match the ground truth simultaneously, higher label cardinality (average 3.08 labels per patch) makes this metric demanding.
- **Dominant Confusion Pair:** *Agriculture* versus *Grassland/Pasture*. In $M1$ and $M2$, exactly 18 patches were falsely classified as Grassland due to spectral similarities between tilled clover meadows and grazed alpine turf.

9.2 The Wetland Classification Failure

Across all three models at $T=0.50$, **Wetland achieved an F1-score of exactly 0.0000**:

- Ground Truth Wetland patches in Test: **27**
- True Positives (TP): **0**
- False Positives (FP): **0**
- False Negatives (FN): **27**

Why Did the Models Fail on Wetland?

1. **Severe Training Imbalance (2.7% Prevalence):** In tile `T33UUP`, the official training split provided **only 8 positive Wetland patches out of 300** (292 negatives, 97.3%).
2. **Loss Formulation Dynamics:** In unweighted Binary Cross-Entropy loss, misclassifying 8 positive patches contributes very little gradient signal compared to avoiding false alarms on the 292 negative patches. Consequently, the network learned that outputting strongly negative logits for Wetland ($z_{\text{wetland}} \in [-3.5, -0.8]$) minimized overall loss.
3. **Threshold Calibration Insight:** When tuning decision thresholds on the validation set, lowering the threshold in M_3 to $T=0.20$ yielded non-zero validation recall, but on the test set it introduced false alarms in damp agricultural fields.

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10. Conclusions and Limitations

10.1 Conclusions

1. **Multi-label recognition is essential:** Over 82% of Sentinel-2 patches in European landscapes contain co-occurring land covers.
2. **NIR ($B08$) provides the decisive performance boost:** Adding NIR in M_2 increased default-threshold Macro-F1 by **+11.32 percentage points** (0.5978 to 0.7110), boosted Exact Match from 12% to 38%, and resolved woody scrub confusion (*Shrub* F1 surging from 0.3529 to 0.9114).
3. ***$M2$ performed best on this curated test set:*** *$M3$* achieved 0.5673 Macro-F1, demonstrating that adding SWIR on small sample sizes without pretraining can degrade generalization through overfitting.
4. **Minority classes require dedicated loss weighting:** Extreme training imbalance (8 Wetland patches) causes zero-recall failure under unweighted BCE loss.

10.2 Practical Limitations

1. **Single-Tile Geographic Scope:** All 500 patches originate from Tile `T33UUP` in the Austrian Central Alps. The findings reflect Central European temperate alpine geography and cannot be claimed to generalize to Mediterranean, Scandinavian, or arid biomes.
2. **Intentional Class-Selection Bias:** To enable evaluation of rare classes, we collected 100% of the tile's Wetland and Shrub patches. This enriched Wetland to 10.2% and Shrub to 22.8% across the 500 patches, diverging from raw landscape prevalence (1.6% and 3.6%).

10.3 Future Improvements

- **Positive-Class Weighting / Focal Loss:** Apply positive class weights ($w{\text{pos}} = N{\text{neg}} / N_{\text{pos}}$) to penalize minority-class omissions.
- **Multispectral Transfer Learning:** Pretrain the CNN backbone on continental-scale archives before fine-tuning on regional tiles.
- **Guided Super-Resolution:** Downscale SWIR bands using 10 m visible band spatial guidance rather than unguided bilinear interpolation.

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11. References

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